

Yongxiang Lei

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Experience

- | | |
|---|-------------------|
| Senior Technical Scientist , ClearMotion – Billerica, MA | 11/2023 — Present |
| <ul style="list-style-type: none">• Leading R&D activities for motion planning and graph-based SLAM involving road surface data• Leading R&D activities for in-vehicle entertainment features with active suspension systems• Leading applied research activities with partners in both academia and industry• Managing the graduate student research intern program in the CTO office | |
| Principal Scientist , ClearMotion – Billerica, MA | 05/2021 — 11/2023 |
| <ul style="list-style-type: none">• Responsible for Z-motion planning and control with active suspension | |
| Owner and Consultant , Findop Solutions LLC – Wellesley, MA | 04/2020 — 04/2021 |
| <ul style="list-style-type: none">• Technical consulting on autonomous driving for startup companies | |
| Technical Lead – Control Systems , ISEE AI – Cambridge, MA | 10/2019 — 04/2020 |
| <ul style="list-style-type: none">• Led the controls team of five engineers• Designed planning and control algorithms for robust and accurate trailer backing• Led the cross-team efforts on designing the planning and control interface | |
| Principal Controls Engineer , ISEE AI – Cambridge, MA | 01/2019 — 10/2019 |
| <ul style="list-style-type: none">• Started the controls team• Improved the steering and speed control algorithms for passenger and commercial vehicles• Integrated MBD (model-based design) into autonomous software development• Supported weekly road testing and data analysis | |
| Senior Controls Researcher , ClearMotion – Billerica, MA | 07/2017 — 01/2019 |
| <ul style="list-style-type: none">• Responsible for R&D of preview suspension control | |
| Software Engineer , MathWorks – Natick, MA | 07/2015 — 06/2017 |
| <ul style="list-style-type: none">• Core developer of Simulink Control Design Toolbox• Led cross-team collaboration with Simulink Solvers and Physical Modeling | |
| Application Support Engineer , MathWorks – Natick, MA | 06/2014 — 06/2015 |
| <ul style="list-style-type: none">• Technical support in the CDA (Control Design and Automation) group• Worked with Stateflow and Simulink Solvers Team on feature development and software testing | |
| Research Intern , Mitsubishi Electric Research Laboratories – Cambridge, MA | 06/2013 — 08/2013 |
| <ul style="list-style-type: none">• Summer intern in the Mechatronics Group• Solved a non-convex optimization problem in the optimal control of an electromechanical system. | |
| Research and Teaching Assistant , New York University – Brooklyn, NY | 06/2013 — 08/2013 |
| <ul style="list-style-type: none">• Researched on the integration of nonlinear control theory with reinforcement learning.• Researched on mathematical modeling of the central nervous systems (CNS) and physiological feedback control mechanisms.• Teaching Assistant in the lab session of an undergraduate feedback control class | |
| Research Assistant , The Chinese University of Hong Kong – Hong Kong | 10/2007 — 12/2007 |
| <ul style="list-style-type: none">• Worked in the Dept. of Mechanical and Automation Engineering• Researched on output regulation theory for non-minimum phase nonlinear systems | |

Education

Politecnico di Milano, PhD in Mechanical Engineering (Control) 10/2020 – 06/2024

- Top 1 in the ECE Department PhD Qualifying Exam
- Best PhD Dissertation award in the ECE Department
- Minor in Mathematics at Courant Institute of Mathematics
- Dissertation topic was on learning-based soft sensing and prediction for industrial applications
- Dissertation has been published as a book with Elsevier in 2025

Central South University, MS in Control Engineering 09/2017 – 06/2020

- GPA 88/100, ranked top 5% of department
- Visited the Hong Kong Chinese University to work on nonlinear control

University of South China, BS in Automation 09/2013 – 06/2017

- Selected for National Key Class, a program for students achieving top scores in the national college entrance exam
- GPA 85/100, ranked top 10% of class

Publications

Book

- [1] Y. Jiang and Z.-P. Jiang, *Robust Adaptive Dynamic Programming*. John Wiley & Sons, 2017.

Book Chapters

- [1] S. Guo, Y. Jiang, J. Li, D. Zhou, S. Su, M. J. Bocus, X. Zhu, Q. Chen, and R. Fan, “Road environment perception for safe and comfortable driving,” in *Autonomous Driving Perception: Fundamentals and Applications*. Springer, 2023, ch. 11, pp. 357–387.
- [2] Y. Jiang and Z.-P. Jiang, “Robust adaptive dynamic programming,” in *Reinforcement Learning and Approximate Dynamic Programming for Feedback Control*. Elsevier, 2012, ch. 13, pp. 281–302.

Journal Papers

- [1] Y. Lei, “A wave forecasting method based on probabilistic diffusion lstm network for model predictive control of wave energy converters,” *Applied Soft Computing*, vol. 164, p. 112 006, 2024.
- [2] Y. Lei, “Intelligent optimal framework for the industrial mining plant-wide prediction control,” *IEEE Transactions on Instrumentation and Measurement*, vol. 72, pp. 1–12, 2023.
- [3] Y. Lei and H. R. Karimi, “Dualstm: A novel key-quality prediction for a hierarchical cone thickener,” *Control Engineering Practice*, vol. 137, p. 105 566, 2023.
- [4] Y. Lei and H. R. Karimi, “Underflow concentration prediction based on improved dual bidirectional lstm for hierarchical cone thickener system,” *The International Journal of Advanced Manufacturing Technology*, vol. 127, no. 3, pp. 1651–1662, 2023.
- [5] Y. Lei, X. Chen, Y. Xie, and L. Cen, “A self-supervised temporal temperature prediction method based on dilated contrastive learning,” *Journal of Process Control*, vol. 120, pp. 150–158, 2022.
- [6] Y. Lei and H. R. Karimi, “A digital twin model of three-dimensional shading for simulation of the ironmaking process,” *Machines*, vol. 10, no. 12, p. 1122, 2022.
- [7] Y. Lei, H. R. Karimi, and X. Chen, “A novel self-supervised deep lstm network for industrial temperature prediction in aluminum processes application,” *Neurocomputing*, vol. 502, pp. 177–185, 2022.
- [8] Y. Lei, F. Liu, H. R. Karimi, and X. Chen, “Manifold semi-supervised learning for aluminum electrolysis temperature identification based on regularized hierarchical extreme learning machine,” *Proceedings of the Institution of Mechanical Engineers, Part I: Journal of Systems and Control Engineering*, vol. 236, no. 6, pp. 1109–1118, 2022.

- [9] N. Long, Y. **Lei**, L. Peng, P. Xu, P. Mao, *et al.*, “A scoping review on monitoring mental health using smart wearable devices,” *Math. Biosci. Eng.*, vol. 19, no. 8, pp. 7899–7919, 2022.
- [10] Y. **Lei**, X. Chen, and Y. Xie, “An improved cell situation identification approach with convolutional neural network and wavelet extreme learning machine,” *Proceedings of the Institution of Mechanical Engineers, Part I: Journal of Systems and Control Engineering*, vol. 235, no. 10, pp. 1898–1905, 2021.
- [11] Y. **Lei**, H. R. Karimi, L. Cen, X. Chen, and Y. Xie, “Processes soft modeling based on stacked autoencoders and wavelet extreme learning machine for aluminum plant-wide application,” *Control Engineering Practice*, vol. 108, p. 104706, 2021.
- [12] Y. **Lei**, X. Chen, M. Min, and Y. Xie, “A semi-supervised laplacian extreme learning machine and feature fusion with cnn for industrial superheat identification,” *Neurocomputing*, vol. 381, pp. 186–195, 2020.
- [13] M. Min, X. Chen, Y. **Lei**, Z. Chen, and Y. Xie, “A novel kernel-based extreme learning machine with incremental hidden layer nodes,” *IFAC-PapersOnLine*, vol. 53, no. 2, pp. 11836–11841, 2020.
- [14] Y. **Lei**, L. Cen, X. Chen, and Y. Xie, “A hybrid regularization semi-supervised extreme learning machine method and its application,” *IEEE Access*, vol. 7, pp. 30102–30111, 2019.

Conference Papers

- [1] Y. **Lei**, H. Karimi, *et al.*, “A self-supervised lstm network for cell temperature prediction in aluminium electrolysis reduction,” in *18th International Conference on Condition Monitoring and Asset Management, CM 2022*, British Institute of Non-Destructive Testing, 2022, pp. 259–266.
- [2] Y. Jiang, P. Li, and Y. **Lei**, “A novel superheat identification of aluminum electrolysis with kernel semi-supervised extreme learning machine,” in *Journal of Physics: Conference Series*, IOP Publishing, vol. 1631, 2020, p. 012005.
- [3] Y. **Lei**, X. Chen, and W. Gui, “Hessian regularization semi-supervised extreme learning machine for superheat identification in aluminum reduction cell,” in *2019 Chinese Control And Decision Conference (CCDC)*, IEEE, 2019, pp. 4406–4411.

Patents

- [1] C. Guan and Y. **Jiang**, *Simultaneous localization and mapping using road surface data*, US Patent 11,933,616, Mar. 2024.
- [2] C. Guan and Y. **Jiang**, *Vehicle sensors arrangement and method for mapping the road profiles*, US Patent 11,884,278, Jan. 2024.
- [3] V. Sridhar, J. P. Eisenmann, Y. **Jiang**, and D. F. Weatherwax, *Road surface-based vehicle control*, US Patent App. 18/344,943, Mar. 2024.
- [4] M. Giovanardi, M. F. Alanis, J. A. Ekchian, P. A. DiZio, J. R. Lackner, A. C. Chetty, and Y. **Jiang**, *Method and apparatus for motion sickness mitigation in a vehicle*, US Patent App. 17/920,642, May 2023.
- [5] V. Sridhar, J. P. Eisenmann, Y. **Jiang**, and D. F. Weatherwax, *Road surface-based vehicle control*, US Patent 11,733,707, Aug. 2023.
- [6] F. Zhang, Y. **Jiang**, and M. K. Yeddanapudi, *Systems and methods for highlighting graphical models*, US Patent 11,853,690, Dec. 2023.
- [7] A. C.-H. Chen, Y. **Jiang**, J. A. Ekchian, M. Giovanardi, *et al.*, *Vehicle localization systems and methods*, US Patent App. 17/489,924, Jun. 2022.
- [8] J. P. Eisenmann, W. Graves, Y. **Jiang**, M. J. Proctor, and J. A. Ekchian, *Multi-lane road characterization and tracking algorithms*, US Patent App. 17/773,676, Dec. 2022.
- [9] Y. **Jiang**, V. Sridhar, J. P. Eisenmann, W. Graves, M. W. Finnegan, *et al.*, *Vehicular localization systems, methods, and controls*, US Patent App. 17/436,012, Jun. 2022.
- [10] V. Sridhar, J. P. Eisenmann, Y. **Jiang**, and D. F. Weatherwax, *Road surface-based vehicle control*, US Patent 10,901,432, Jan. 2021.
- [11] V. Sridhar, J. P. Eisenmann, Y. **Jiang**, and D. F. Weatherwax, *Vehicle control based on localization and road data*, US Patent US11801726B2, May 2020.
- [12] Y. Wang, Y. **Jiang**, and B. A. Scott, *Motion-control system for performing different tasks*, US Patent US9529341B2, Dec. 2016.

Awards

Excellent Reviewer	2016
• Journal of Guidance, Control, and Dynamics (JGCD), Oct. 1st, 2015 – Sept. 30th, 2016.	
Alexander Hessel Award	2015
• Best Ph.D. Dissertation in Electrical Engineering, ECE Dept in Tandon School of Engineering at NYU, 2015	
Shimemura Young Author Prize	2013
• The 9th Asian Control Conference, Istanbul, Turkey, June 2013.	
Richard Rosenthal Award	2010
• Outstanding (Top 1) performance on the electrical engineering PhD Qualifying Examination, ECE Dept in Tandon School of Engineering at NYU, 2010	
Huawei Scholarship for outstanding students	2008
• South China University of Technology, 2008	
National First Place	2007
• The Chinese Graduate Mathematical Contest in Modeling, 2007	
National First Place	2005
• The Chinese Undergraduate Mathematical Contest in Modeling, 2005	

Services

Reviewer for Conferences and Journals

- Conferences: American Control Conference; Conferences on Decision and Control
- Journals: IEEE Transactions on Automatic Control (TAC); Automatica; IEEE Transactions on Neural Network and Learning Systems (TNNLS)

Early Career Advisory Board

2022 – Present

- Control Engineering Practice (CEP), an IFAC Journal